Felipe Campoverde

Robotics Software Engineer

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EDUCATION

Virginia Tech B.S. Computer Science · GPA 3.09 (Major 3.33) · Blacksburg, VA *Expected Dec 2026*

EXPERIENCE

Founding Engineer · Daily's (Imitation Learning, VLAs, ML for Robotics, ROS 2, PyTorch, Docker) *Dec 2025 – June 2026*

- Engineered the full robotics software and learning stack of an autonomous kitchen cell built around a bimanual **OpenArm** robot (dual 7-DOF arms, 14 DOF) that retrieves, cooks, and assembles food bowls across an integrated oven, warmer, and fridge for delivery fulfillment.
- Built an end-to-end imitation-learning pipeline: VR teleoperation (Meta Quest 3 controllers streamed over ADB/TCP to the Jetson) for demonstration capture, **LeRobot**-format data packaging, and cloud training on **Modal** B200 GPUs.
- Implemented a **500 Hz real-time control loop** (ROS 2 ros2_control) driving 14 **Damiao motors** over CAN bus with gravity compensation, behind a 6-layer safety system (e-stop, hardware limits, per-cycle delta guards, collision avoidance, and temperature monitoring).
- Trained and benchmarked multiple vision-language-action policies including **ACT, SmolVLA, π0, π0-FAST, and π0.5 (4B)**, running parameter sweeps and implementing Real-Time Chunking for ~2x inference speedup, plus DAgger and action-smoothing techniques to improve responsiveness and precision.
- Ran the full edge stack on the **Jetson Thor**, covering system services, model inference (PyTorch) with **inverse kinematics in PyBullet**, robot control, and coordination of the RP2350 Picos and Raspberry Pi, all tied together by **ROS 2 (Jazzy)** with ZED stereo + multi-camera perception; shipped a working demo via deterministic trajectory replay after evaluating VLA readiness.
- Containerized the platform with **Docker** integrated with GitHub for distribution and one-step deployment, splitting the stack into two images: one for teleoperation control and dashboard visualization, one for streamlined deployment to the Jetson.

Chief Engineer, CroQuest · VT CRO (C/C++, ESP32, BLE, PlatformIO) *Mar 2025 – Jan 2026*

- Led embedded software development for a custom educational handheld game console (ESP32-WROOM, LCD display, SD card, buttons, D-pad) built for Virginia Tech summer camps to teach kids embedded systems and game development.
- Wrote 5,000+ lines of C/C++ to ship 8+ retro-style games including Snake, Tetris, and Tic-Tac-Toe, plus a 2-player Bluetooth-multiplayer Pong and a Mario-inspired platformer with gravity, momentum, and a scrolling camera.
- Implemented core firmware integrating the LCD, SD storage, and input hardware with asset rendering and UI feedback, plus an SD-card level system letting students author and load their own custom levels.

PROJECTS

Reinforcement Learning Capstone · Virginia Tech (Python, NumPy, Matplotlib, pytest, SciPy) *2025*

- Implemented and benchmarked **Q-Learning, SARSA, and Dyna-Q** in a custom stochastic GridWorld environment (walls, pits, and wind noise) to compare learning speed and policy reliability.
- Ran a planning-step sweep on Dyna-Q and analyzed sensitivity to ε-greedy schedules, quantifying how model-based planning improves sample efficiency and convergence.
- Evaluated policy robustness across random seeds, layout variations, and wind-noise levels, with a tested codebase (pytest) and a reproducible notebook pipeline.

LEADERSHIP & TEACHING

Student Assistant Manager · Virginia Tech Dining *Jan 2024 – Jan 2026*

- Supervised 10+ student staff in one of VT's largest dining halls; enforced food-safety and service standards with bilingual (English/Spanish) support.

Cybersecurity Instructor · VT College of Engineering *Jun 2024 – Aug 2024*

- Designed and taught hands-on cybersecurity lessons (encryption, ethical hacking, CTF challenges) to middle-school summer-camp students.

SKILLS

Languages: C, C++, Python, Java, JavaScript, Haskell, Lua, HTML/CSS

Robotics & ML: ROS 2 (ros2_control), PyTorch, VLAs (π0.5 / SmolVLA / ACT), LeRobot, NVIDIA Jetson Thor, CAN bus, Damiao motors, PyBullet (inverse kinematics), Real-Time Chunking, Modal, Docker, VR teleoperation (Meta Quest 3)

Embedded & Systems: ESP32, RP2350 / RP2040 (Pico), PlatformIO, Arduino, BLE, LiDAR, RealSense, Linux (Arch), Git

Web & Full-Stack: React, Node.js, Express, MongoDB, REST APIs, WebGL **Languages spoken:** Bilingual (English / Spanish)